KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "Asymmetric cryptosystems are based on clever application of number theory concepts" – individually establish it for RSA, ElGamal and Paillier cryptosystems about different keys. (12) [CO2]
   b) Establish usages of both block and stream ciphers in cases of two asymmetric and one symmetric cryptosystems. If not possible to deploy, also establish it. (12) [CO3]
   c) Discuss properties of digital signatures. Analyze them about ElGamal digital signature. (11) [CO2]

2. a) Discuss characteristics of cryptographic hash functions. Digital signature, asymmetric cryptosystem and hash function need to combine concurrently – why and how? (13) [CO1]
   b) Define and discuss characteristics of 'Zero Knowledge Proof'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14) [CO2]
   c) Derive a bonding among order, prime, and primitive element considering the prime 13 and order 2. (08) [CO4]

3. a) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11) [CO1]
   b) Discuss the following attacks: (i) Ciphertext-only, (ii) Chosen-plaintext, (iii) DoS, and (iv) DDoS. How Known-plaintext-attack can be mounted on RSA cipher? Using the required parameters, derive the mechanism. (15) [CO4]
   c) Define Firewalls. Discuss major types of Firewalls. Discuss various network threats elaborately. (09) [CO2]

4. a) Describing required conditions and parameters, discuss how RSA and ElGamal cipher can be used to develop product cipher. Derive re-randomization property of ElGamal cipher. (15) [CO3]
   b) 'Confusion' and 'Diffusion' are the ultimate goal of every cryptosystem – why? Discuss Shannon characteristics of good cipher. (09) [CO1]
   c) Establish a comparison between AES and DES. Show that the encryption and decryption operations of Caesar cipher work according to substitution cipher. (11) [CO2]

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define the Discrete Logarithm Problem (DLP) and explain why it is considered computationally difficult to break in cryptography. (09) [CO1]
   b) In the Diffie-Hellman Key Exchange (DHKE) protocol, each participant selects a private key '$a$' from the range $\{2, \dots, P\text{-}2\}$ where $P$ is a prime number. Explain why the values $a = 1$ and $a = P - 1$ are excluded from this range. (11) [CO2]
   c) Explain scalar multiplication operation on Elliptic Curve (EC). Given an EC (15) [CO3]
   $$E: y^2 = x^3 + 3x + 5 \bmod 17$$ and point $P = (1,3), Q = (4,8)$,
   compute $2P = P + P = (x_3, y_3)$ and $P + Q = (x_4, y_4)$. Consider the equations:
   $$x_3 = S^2 - x_1 - x_2 \bmod P$$
   $$y_3 = S(x_1 - x_3) - y_1 \bmod P$$
   $$S = \begin{cases} \dfrac{3x_1^2 + a}{2y_1} \bmod P \\ \dfrac{y_2 - y_1}{x_2 - x_1} \bmod P \end{cases}$$



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6. a) Explain Post-Quantum Cryptography (PQC) and discuss its impact on symmetric and asymmetric cryptographic algorithms. (07)
   b) Explain the DHKE protocol and its key setup. Describe the EC-DHKE protocol and compare it with standard DHKE in terms of security and key size. (13) [C
   c) Define Transport Layer Security (TLS) and its purpose. Describe the TLS 1.2 handshake process. Compare TLS 1.2 and TLS 1.3 in terms of handshake efficiency and latency. Is O-RTT handshake supported in TLS 1.3, and under what conditions? (15) [CO4

7. a) Explain the $N^2$ key distribution problem in symmetric key systems. (05) [CO1]
   b) Define Elliptic Curve Discrete Logarithm Problem (ECDLP) and explain why computing it is considered difficult. (05) [CO2]
   c) Explain Elliptic Curve Cryptography (ECC) by describing its key generation, encryption and decryption processes. (10) [CO3]
   d) Illustrate the following web attacks with diagrams showing attacker-victim-server interactions, and explain effective mitigation techniques for each: (15) [CO4]
      i) Cross-site Request Forgery (CSRF).
      ii) Server-side Request Forgery (SSRF).
      iii) Cross-site Scripting (XSS).

8. a) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (05) [CO2]
   b) Define the Point at Infinity in ECC and explain the conditions under which it is reached. (06) [CO2]
   c) Explain homomorphic encryption and its types. Derive the equations of homomorphic operations in the Paillier cryptosystem and illustrate each with an example. (12) [CO3]
   d) Illustrate a Man-in-the-Middle (MITM) attack on the DHKE protocol and discuss methods to prevent such attacks in public key cryptosystems. (12) [CO3]



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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Computer and Network Security

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define and discuss goals of computer security. Using a block diagram show relation and explain terminologies of any cryptosystem. (08) [CO1]
   b) Discuss Shannon characteristics to describe characteristics of good cipher. "Ciphertext size should not be larger than plaintext" – show how RSA and ElGamal cryptosystems retain this criterion. (14) [CO2]
   c) Define and discuss characteristics of zero-knowledge proof. How and why digital signatures with public key cryptography is required to combine? Explain the procedure. (13) [CO2]

2. a) Using block diagrams show that asymmetric cryptosystem is alluring than symmetric cryptosystem. Illustrate the major steps of each cycle of AES algorithm using an appropriate diagram. (13) [CO1]
   b) Discuss characteristics of cryptographic hash function. Asymmetric cryptosystem and hash technique need to combine concurrently – why and how? (13) [CO1]
   c) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (09) [CO2]

3. a) Discuss properties retained by digital signatures (DS). Derive the steps of ElGamal digital signature. Show that it is capable to satisfy properties of DS. (13) [CO3]
   b) Define the term cryptanalysis. Explain three terms allied with attack on cryptanalysis. (10) [CO1]
   c) Explain the working procedure of Double DES and Triple DES. Establish a comparative analysis between AES and DES based on various features of design. (12) [CO2]

4. a) ElGamal cipher can be developed to form 'product cipher' and 're-randomized cipher'. How? Show step by step. (14) [CO4]
   b) Derive encryption-decryption procedure of Caesar cipher and Vernam cipher. For RSA algorithm from $ed \equiv 1 \bmod \varphi(n)$, establish that many values may exist for $d$ for a single value of $e$, where symbols have their usual meanings. (14) [CO3]
   c) Derive a bonding among order, prime and primitive element considering the prime 13 and order 2. (07) [CO4]

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are covert channels? Differentiate between error and fault. (08) [CO2]
   b) What is 'Salt' in rainbow table attack? How does viruses gain control in file directory? (10) [CO1]
   c) What is segmentation fault in buffer overflow attack? Explain with figure. Also, discuss the countermeasures of buffer overflow attack. (10) [CO2]
   d) How webcams are used in Distributed Denial of Service (DDOS) attack? (07) [CO3]

6. a) Explain how paging removes the drawbacks of segmentation. (08) [CO3]
   b) Show how integer overflow attack compromises a program's reliability and security. (15) [CO4]
   c) How does a Demilitarized Zone (DMZ) network work? "Paillier cryptosystem permits addition of two ciphertexts, or one ciphertext and one plaintext or multiplication of one ciphertext and one plaintext" – justify it. (12) [CO4]

7. a) "Format string vulnerability is responsible to crash a program" – explain it. (05) [CO2]
   b) Explain relocation with example. What are the advantages of segmentation in operating system? (10) [CO2]
   c) Write short notes on the followings: (12) [CO1]
   (i) Trojan horse, (ii) Boot sector viruses, (iii) Dictionary attack.
   d) What is macro? Discuss how does document virus attack work in excel file using Dynamic Data Exchange (DDE). (08) [CO4]



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8. a) What is a virus signature? How does antivirus work on a system? (10) [CO2]
   b) Differentiate between Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). (10) [CO2]
   c) What is light weight cryptography? Give an example and working procedure of it. (10) [CO4]
   d) Describe Stateful Inspection firewalls. (05) [CO1]



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose to withdraw money from your Bank account when you put a signature over your Bank cheque, what are the properties essential to retain concerning it? Explain them. Now, analyze that RSA algorithm is capable to fulfill those properties or not. Establish your intuition relating the specific portion of RSA algorithm. (13)
   b) Differentiate between 'Cryptography' and 'Cryptanalysis'. Justify them, i.e., how they are associated with Caesar cipher technique. (10)
   c) List the trends in case of any good cipher. Analyze relevancy of 'Known-Plaintext-Attack' with RSA encryption algorithm. (12)

2. a) Discuss major goals (i.e., CIA) of security. Using a diagram state and explain ingredients of a cryptosystem. (09)
   b) Explain properties of cryptographic hash function. How and why data encryption and hash techniques need to perform concurrently? Propose an outcome from your thinking. (11)
   c) Discuss pros and cons of 'Stream cipher' and 'block cipher'. Assess the working procedures of Caesar cipher, RSA cipher, Vernam cipher etc., with block and Stream ciphers. (15)

3. a) Relating order, prime and primitive element, demonstrate a bonding. Also, judge your bonding for a prime 13, and order 2. (08)
   b) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11)
   c) Discuss characteristics maintained by "Good Viruses". (06)
   d) Considering two messages $m_1$ and $m_2$, develop a model of ElGamal product cipher. Also demonstrate the decryption technique. Step wise write the whole procedure. (10)

4. a) Design a procedure to develop data encryption and digital signature for RSA algorithm together. (08)
   b) "ElGamal digital signature is based on discrete logarithm"—how discrete logarithm manages key generation, signing, verification, etc.? Explain. (11)
   c) How ElGamal cipher can be modified to re-randomize itself? Explain. (07)
   d) Discuss various aspects that make computer networks vulnerable. (09)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain relocation of address with example. What are the complementary goals to protect object of OS? (09)
   b) Discuss how ECC works based on the basic of trapdoor function. (10)
   c) Explain the types of Salami attack. What are the impacts and preventions of DHCP spoofing attack? (10)
   d) Differentiate among black hat, white hat and grey hat hackers. (06)

6. a) What is polymorphic virus? How does an Email impersonation attack work? (08)
   b) What are the considerations before committing to a hybrid firewall solution? Let, a businessman works with a public website, accessible from the internet, which put the entire internal network at high risk. Explain which network can help him to conduct his business securely and how? (15)
   c) Explain with figure how paged segmentation removes the problems associated with segmentation and paging. (12)



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7. a) Discuss the problems created by format string attack and countermeasures for them. (15)
   b) Demonstrate the ways to keep devices safe from wiretapping. (12)
   c) Which one is preferable, RSA or, ECC? Explain why? (08)

8. a) Suppose an attacker may introduce extra code, sending new instructions to the application to gain access to IT systems. If attackers know the memory layout of a program, they can intentionally overwrite the executable code with their own code. What kind of attack is this? What are the countermeasures for it? (15)
   b) Explain how documents virus works in Microsoft's excel Microsoft's word file using DDE protocol. (10)
   c) What are the responsibilities of firewall administrators? What are the challenges of intrusion detection system? (10)



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the terms: 'Security services', 'Cryptosystem', 'Cryptoanalysis' along with the major goals of security. Also explain a conventional cryptosystem by using a proper diagram. (13)
   b) Specify Shannon characteristics to explain good ciphers along with 'confusion' and 'diffusion'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14)
   c) Discuss the benefits of an asymmetric cryptosystem. For RSA algorithm from $ed \equiv 1 \; mod \; \varphi(n)$ show that for a single value of $e$, there may exist many values of $d$, where symbols have their usual meanings. (08)

2. a) Define chosen-plaintext attack. Show how RSA algorithm suffers with this attack. (08)
   b) Explain the working procedure of Caesar cipher. How it is an example of substitution ciphers. (07)
   c) Discuss the properties of digital signatures. RSA algorithm can be used to implement product cipher. How? Show it? (10)
   d) Discuss the techniques, pros and cons of block and stream ciphers. What are the trends of maintaining security? Explain them. (10)

3. a) Explain the working procedure of ElGamal encryption algorithm. Why it is said to be a probabilistic encryption algorithm? Explain. (13)
   b) Explain ElGamal digital signature algorithm in detail. Also give a working example of it. (13)
   c) Discuss the major characteristics of cryptographic hash function? Why it is one-way? (09)

4. a) What is firewalls? Discuss the characteristics of various malicious code. (09)
   b) Explain AES algorithm with a block diagram. Establish a comparative study between AES and DES. (14)
   c) Discuss the characteristics of good viruses. (05)
   d) How triple DES is better than DES algorithm? Explain primitive element of Elgamal cryptosystem. (07)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is segmentation fault? Explain Buffer Overflow attack with proper code and stack diagram. (15)
   b) Show how Integer Overflow attack compromises a program's reliability and security. (15)
   c) Explain document virus using Microsoft's Dynamic Data Exchange (DDE) protocol. (05)

6. a) What are the problems associated with Segmentation and Paging? Explain how Paged Segmentation removes them. (15)
   b) Explain some countermeasures that are used against Buffer Overflow attack. (15)
   c) What is format string? (05)

7. a) Explain attacks on format string vulnerability. (15)
   b) Write short notes on the followings: (12)
      (i) DHCP Spoofing Attack. (ii) Rainbow Table Attack.
      (iii) Polymorphic Viruses. (iv) Dictionary Attack.
   c) What are covert channels? Differentiate between errors and faults. (08)

8. a) How does viruses attach with original programs? (10)
   b) Explain the working principle of Elliptic Curve Cryptography (ECC). What are the reasons to prefer ECC over RSA? (10)
   c) What is salt? How does it help against Rainbow Table attack and make password hashing more secure? (10)
   d) Explain SQL injection and web bug. (05)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer ANY THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer ANY THREE questions from this section in Script A)

1. a) Define the terms: 'confusion', 'diffusion', 'cryptography' and 'cryptanalysis'. Discuss Shannon characteristics of good cipher. (12)
   b) 'RSA is a commutative like cryptosystem' – how? Explain deliberately. (08)
   c) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   d) Define 'known-plaintext' attack. How this attack can mount over RSA? (07)

2. a) Discuss characteristics of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Digital signature and public key cryptography can be combined – why and how? Explain. (08)
   c) Using an example explain ElGamal cryptosystem. How does it relate with discrete logarithm problem? (10)
   d) Discuss threats to e-mail. (05)

3. a) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   b) Explain AES algorithm using a block diagram. Make a comparison between DES and AES. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (09)

4. a) What is meant by homomorphic encryption? Show that RSA and ElGamal cryptosystems are homomorphic. (10)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) Explain the mechanism of triple DES algorithm. (05)

## SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) What is salt and how does it make password hashing more secure? Also explain rainbow table attack and how salt can help against rainbow table attack. (15)
   b) How does the attacker know what algorithm and salt to use in a dictionary attack? (06)
   c) Explain Buffer Overflow attack with proper code and stack diagram. Also discuss about the countermeasures that are used against Buffer Overflow attack. (14)

6. a) Explain how Format String attack works with appropriate activation record diagrams. Also explain different uses of Format String attack with proper example(s) and stack diagrams. (15)
   b) Demonstrate Integer Overflow attack using suitable example(s). (07)
   c) Explain SQL Injection attack with practical example. Also discuss about the prevention techniques used against SQL Injection attack. (13)

7. a) Explain Persistent, Reflected and DOM-based Cross-Site Scripting (XSS) attacks with example(s). (15)
   b) Explain the methods of preventing XSS attacks. (10)
   c) What are web bugs? Explain how attackers bypass CAPTCHA protection. (10)

8. a) Explain how combining Paging with Segmentation provides better performance for protection in general purpose operating system compared to using them separately. (09)
   b) What is a Covert Channel? How to create Covert Channels? (06)
   c) Briefly explain the following attacks: (i) Ransomware, (ii) Salami Attack, (iii) Tapjacking, (iv) Document virus using Microsoft's Dynamic Data Exchange. (16)
   d) What is Access Control Directory? (04)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Through Shannon characteristics, discuss the characteristics of good cipher. Define the terms: 'confusion' and 'diffusion'. (10)
   b) Define known-plaintext attack. How it can be associated with RSA cryptosystem? (10)
   c) On same modulus using RSA, how multiple parties can encrypt data? (09)
   d) Show that "RSA cryptosystem is homomorphic". (06)

2. a) Discuss properties of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Explain the working procedure of ElGamal cryptosystem specifying that it is based on discrete logarithm problem. (10)
   c) Make a comparison between the mechanism of AES and DES. Describe AES algorithm using a block diagram. (13)

3. a) Define cryptographic hash function. Discuss properties of it. (07)
   b) How and why you combine digital signature and public key cryptography? Explain. (08)
   c) Discuss the parameters that make a computer network vulnerable. (10)
   d) Define sensitive data. Discuss the factors that can make data sensitive. (10)

4. a) How documents can be signed using public key cryptography and one way hash function? Discuss. (07)
   b) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   c) How 'primitive element' relates with ElGamal? (08)
   d) What are the requirements of database security? Explain them. (07)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Give an intuitive explanation of the working principle of Elliptic Curve Cryptography (ECC). Also explain how encryption and decryption work in ECC. (13)
   b) Explain ECC based digital signature with mathematical proof. (10)
   c) What are the reasons to prefer ECC over RSA? (05)
   d) What is tapjacking? Explain the prevention mechanism used against tapjacking in context of Android operating system. (07)

6. a) Write short notes on following attacks: (16)
      i. Page-in-the-Middle. ii. Program Download Substitution.
      iii. SQL Injection. iv. Man-in-the-Browser.
   b) Explain how attackers use User-in-the-Middle attack to bypass CAPTCHA protection. (06)
   c) What are web bugs? Explain how websites can be protected against unwanted change both from administrator's and users' perspective. (13)

7. a) With appropriate code and stack diagram explain how Buffer Overflow can be exploited to execute attacker's provided code. Also explain some common counter measures that are used against Buffer Overflow attack. (18)
   b) Explain the uses of Format String attack with appropriate example(s) and stack diagrams. (11)
   c) Explain Integer Overflow attack using suitable example(s). (06)

8. a) Write short notes on the followings: (14)
      i. Salami Attack. ii. Covert Channel.
      iii. Document Virus using Microsoft's Dynamic Data Exchange (DDE).
   b) Explain the problems associated with Access Control Directory. (06)
   c) Explain the problems associated with Segmentation and Paging. Also discuss how combining Paging with Segmentation provides better performance for protection in general purpose operating system. (15)

CR CSE 14

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   b) Through Shannon characteristics, discuss the characteristics of good cipher. Compare between block and stream ciphers. (10)
   c) Define chosen-plaintext attack. Discuss a specific key exchange protocol to distribute a key. (10)
   d) Define cryptographic hash function. Discuss some properties of it. (07)

2. a) Explain the mechanism of triple DES encryption. Using a block diagram explain AES algorithm. (13)
   b) Discuss properties of digital signature. Explain Elgamal digital signature algorithm with an example. (12)
   c) A computer network can become vulnerable. How? Explain the reasons. (10)

3. a) In RSA, how $ed = 1$? On same modulus using RSA, how multiple parties can encrypt data? (10)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Explain why and how digital signature and public key cryptography can be combined? (08)
   d) Explain the steps of signing documents with public key cryptography and one way hash functions. (07)

4. a) Discuss the dimension of reliability and integrity of database. What are the requirements of database security? Explain them. (10)
   b) Define 'primitive element' with example. How it is associated with Elgamal cryptosystem? (10)
   c) Define 'confusion' and 'diffusion'. Give an overview of DES encryption technique. (08)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (07)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Consider the code snippet shown in below: (07)

```
int main()
{
  int pass = 0;
  char buff[3];
  printf("\n Enter the password:\n");
  gets(buff);
  if(strcmp(buff,"cse"))
  {
   printf("\n Wrong password\n");
  }
  else
  {
   printf("\n Correct password\n");
   pass = 1;
  }
if(pass)
 {
  /*Give admin priviledge to the  user*/
  printf("\n Admin priviledge given to the user\n");
  }
  return 0;
}
```

Explain how buffer overflow attack can be utilized to get admin privilege without even knowing the actual password.



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b) With appropriate stack layout explain how buffer overflow attack can be used to modify return address of a function. Design an input payload for the code snnipet shown in the following to transfer the execution to secret function. (15)

```
void secretFunction()
{
printf("Can't be called");
}
void test()
{
char buff[10];
gets(buff);
}
int main()
{
test();
return 0;
}
```

```
Dump of assembler code for function
secretFunction:
0x004014d0<+0>: push %ebp
0x004014d1<+1>: mov %esp, %ebp
.
.
.
```

c) Explain how buffer overflow can be exploited to execute attacker's provided code. (06)
d) Explain integer overflow attack with an example. (07)

6. a) Discuss some prevention mechanisms that are used against buffer overflow attack. (12)
b) Explain why the code snnipet shown in the below is vulnerable to format string attack. (06)

```
int main (int argc, char *argv[])
{
char *secret = "test";
char *secret2 = "test2";
printf(argv[1]);
return 0;
}
```

c) Explain how format string vulnerability can be utilized for: (i) Crashing a program (ii) Viewing the stack (iii) Viewing memory at any location (iv) Writing an integer to nearly any location in process memory. (17)

7. a) Suppose, you want to perform brute force attack on a system which allows password length between 5 to 8 characters limited to character set:{'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', '1', '2', '3', '4'}. Now, write a pseudocode to generate all possible passwords for your brute force attack. (12)
b) Explain how the tapjacking attack works. (05)
c) What are covert channels? Explain how covert channels can be created. (12)
d) Write short notes on: (i) Trapdoors (ii) Keystroke logging (06)

8. a) Explain how Dynamic Data Exchange (DDE) and macro in Microsoft Office can be exploited to run malicious code. (05)
b) Explain the working principle of ransomware. (05)
c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed DoS attack. (08)
d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
e) What is meant by malleability? Show that both RSA and Elgamal are malleable cryptosystem. (10)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2016
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: cryptography, cryptanalysis. Discuss Shannon characteristics of good cipher. (10)
   b) Define known-plaintext attack. How can it be applied for RSA cryptosystem? (10)
   c) Explain with example: a) substitution ciphers, b) transposition ciphers. Discuss a specific key exchange protocol to distribute a key. (10)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (05)

2. a) What are the characteristics of digital signature? Explain ElGamal digital signature algorithm with an example. (12)
   b) Define cryptographic hash function. Discuss some properties of it. (08)
   c) Explain how $ed = 1$ in RSA encryption algorithm? [Where $d$ and $e$ have their usual meanings]. (07)
   d) Digital signature and public key cryptography can be combined. Why and how? Explain. (08)

3. a) Discuss the steps to sign documents with public-key cryptography and one way hash functions. (07)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) What are the limitations of RSA cryptosystem? Make a comparison between RSA and ElGamal cryptosystems. (10)
   d) Explain the terms: (i) denial-of-service (DOS) attack and (ii) distributed denial-of-service (DDOS) attack. (08)

4. a) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) What is meant by homomorphic encryption? Explain with examples. (08)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) If two users share access to a segment, they must do so by the same name. Must their protection rights to it be the same? Why or why not? (05)
   b) Why is Kerberos a credible security solution? Describe, how is authentication and access authorization implemented in Kerberos. (10)
   c) Make short notes over (i) Loose-lipped systems (ii) Access control matrix (08)
   d) Briefly explain the operational modes of Intrusion detection system. (12)

6. a) What do you mean by Wired Equivalent Privacy (WEP)? Briefly explain all the possible attacks on wifi protected access (WPA & WPA2) networks. (09)
   b) Draw and explain each of the constructing components of wireless data frame. (09)
   c) Suppose Alice is sending an Email to Bob, but Alice wants her Email to be anonymous. How can Alice achieve this? What are the threats about which Alice is afraid of? (12)
   d) Describe how RSA SecurID suite works. (05)

7. a) Suppose, while browsing web, your machine got attacked by a virus. What are the best places for that virus to make itself hidden and gain control over your system? Describe the situation before and after this infection. (09)
   b) What are the error detecting codes that can be used to detect when digital objects are surreptitiously altered? (10)



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c) Suppose, an attacker wants to attack a target server which has higher bandwidth than the attacker. Explain an attack with necessary figure by which the attacker can overflow the target server's maximum connection capacity. (10)

d) Write down a short note on Smurf Attack. (06)

8. a) What do you mean by security through obscurity and challenge-response system? Briefly explain. (10)

b) How would you distinguish a risk from other project events? Discuss the strategies for dealing with risks. (10)

c) What type of attacks "Keystroke loggers" is? Briefly explain. (08)

d) Suppose, a system is being accessed by different users with different privileges. What are the principles that you as an administrator would adopt while giving access to the users. (07)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2013
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: 'Ciphertext-only attack', 'Known-plaintext attack' and 'Chosen-plaintext attack'. Discuss Shannon characteristics of good cipher. (12)
   b) Explain signing and verification mechanisms of ElGamal Digital Signature scheme with an example. (13)
   c) Discuss how can a computer network become vulnerable? Briefly explain the reasons. (10)

2. a) Discuss the terms: 'Stream ciphers' and 'Block ciphers'. Discuss a specific key exchange protocol to distribute a key. (12)
   b) Discuss the characteristics of relation between the two keys of public key cryptography. Explain how RSA keys are generated employing extended Euclid's gcd algorithm. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (10)

3. a) Discuss the basic protocol of signing documents with public-key cryptography and one-way hash function. (08)
   b) ElGamal encryption mechanism is non-deterministic, how? Explain with a suitable example. (12)
   c) Explain the steps of AEs algorithm with a block diagram. (08)
   d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)

4. a) Define cryptographic hash function. Discuss the properties of it. (08)
   b) Explain the mechanism of triple DES encryption. (07)
   c) Explain the homomorphic encryption mechanism with possible merits and demerits. (10)
   d) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is information security? "Security measures must balance between security and productivity"- justify. (12)
   b) Define risk and risk analysis. Discuss the steps of risk analysis. (11)
   c) Write notes on- i) Confidentiality, ii) Integrity, and iii) Availability (12)

6. a) Differentiate between cryptography and cryptanalysis. Encrypt the following message using column transposition method with five column and Vernam method. (15)
   "CAESAR WAS KILLED BY BRUTUS"
   b) What is the basic theme of OS security? What is fence register? How does fence register provide extra benefit in relocation method? Explain. (12)
   c) What are the differences between paging and segmentation? (08)

7. a) What are malicious error and non-malicious error? Discuss six categories of non-malicious errors. (10)
   b) Explain risk exposure and risk leverage with example. (10)
   c) Discuss exhaustive attack and one time password with an example. Differentiate between copyright, patents, and trade secret. (15)

8. a) Differentiate between laws and ethics. Why is computer crime hard to define and hard to prosecute? Explain. (10)
   b) Explain virus signature with an example. Differentiate between virus and worm. (10)
   c) Explain buffer overflow with an example. How does buffer overflow affect code and data? Discuss with necessary figures. (15)



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2012
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss the ingredients of a cryptosystem briefly. (08)
   b) Discuss the characteristics of asymmetric cryptosystem. Make a comparison between block and stream ciphers. (10)
   c) Discuss the basic protocol of public-key algorithm so that it can be used for digital signatures. Does it satisfy the characteristics of signature? (10)
   d) Why and how one-way hash function is associated with digital signature? (07)

2. a) How RSA keys are generated and what are the limitations of RSA encryption algorithm? Explain with example. (10)
   b) Define 'primitive element' with example. How is it related to ElGamal cryptosystem? (10)
   c) What is meant by malleability? Show that RSA and ElGamal are malleable cryptosystem. (10)
   d) Explain triple DES mechanism. (05)

3. a) Define cryptographic hash function. Discuss some properties of it. (08)
   b) How ElGamal is a probabilistic cryptosystem? Explain. (08)
   c) Define symmetric key cryptosystem. Explain the steps of AES encryption algorithm. (12)
   d) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (07)

4. a) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (08)
   b) How is trade secret protection vanished by 'reverse engineering'? Explain. (08)
   c) Discuss the dimension of reliability and integrity of database. What are the requirements for database security? Explain them. (11)
   d) Define sensitive data. Discuss the steps of interaction among a user, a trusted front end and a DBMS. (08)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the parameters that can make a computer network vulnerable? Discuss them briefly. (10)
   b) What do you mean by security and computer security? "Security measures must balance between security and productivity"- Justify the statement. (10)
   c) Discuss the threats to electronic mail. (07)
   d) Define risk analysis. Discuss the steps of risk analysis. (08)

6. a) Discuss about risk exposure and risk leverage. (10)
   b) How can fence register be used to provide extra benefit in relocation method? What are the advantages of two pair base and bound register? (13)
   c) Discuss tagged architecture with pictorial view. (12)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How does buffer overflow affect code and data? Discuss elaborately. (15)
   c) What is computer virus? Differentiate virus and worm. How is virus attached? (10)

8. a) How does virus gain control? Discuss with pictorial view. Differentiate copyright, patents, and trade secret. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (10)
   c) Differentiate laws and ethics. "Computer crime is hard to define and hard to prosecute"- Justify the statement. (15)



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2011
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define digital signature. Write some properties of digital signature. (09)
   b) How RSA algorithm is used for digital signing? Explain. (08)
   c) Define the terms: cryptosystem, cryptanalysis. Discuss the benefits of asymmetric cryptosystem. (10)
   d) What is meant by homomorphic encryption? Explain with an example. (08)

2. a) Explain ElGamal encryption algorithm with an example. (13)
   b) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (09)
   c) Explain AES encryption algorithm. (09)
   d) Give an example of transposition cipher. (04)

3. a) In RSA algorithm, how $ed = 1$? [ Where $e$ and $d$ have their usual meanings]. (07)
   b) What are the goals of security? Explain them briefly. (06)
   c) What are the parameters that can make a computer network vulnerable? Discuss them. (13)
   d) Discuss the requirements for database security. (09)

4. a) What is the purpose of encryption in a multilevel secure database management system? (08)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (09)
   d) What is cryptographic checksum? Explain with example. (06)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) "Security measures must balance between security and productivity"- Justify the statement. (08)
   b) Define risk analysis. Discuss the steps of risk analysis. (10)
   c) What is security planning? Write down the issues of security planning. (10)
   d) How does operating system offer protection at several levels? (07)

6. a) ~~What is~~ How virus gain control? Discuss with pictorial view. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (12)
   c) "Computer crime is hard to define and hard to prosecute"- Justify. (13)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How buffer overflow affects code and data? Discuss elaborately. (15)
   c) What is Computer Virus? Differentiate between Virus and Worm. How Virus is attached? (10)

8. a) Write short notes on- (i) Risk exposure (ii) Risk leverage. (10)
   b) How fence register can be used to provide extra benefit in relocation method? What is the advantage of two pair base and bound register? (12)
   c) Discuss the tagged architecture with pictorial view. (13)



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2010
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

## SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is public key encryption? How does it differ from private key encryption? (10)
   b) Define homomorphic encryption. RSA and ElGamal cryptosystems are homomorphic – explain. (12)
   c) Why is RSA encryption algorithm deterministic? In RSA, how $ed=1$? [Where $e$ and $d$ have their usual meanings]. (13)

2. a) ElGamal encryption algorithm is probabilistic – explain. (11)
   b) Explain the steps of AES encryption algorithm. (12)
   c) Define hash function. What are the applications of hash function in cryptography? (07)
   d) Explain triple DES mechanism. (05)

3. a) What are the requirements for the security of database? Explain them precisely. (10)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) What are the parameters that can make a network vulnerable? Among them, explain 'anonymity' and 'sharing'. (13)

4. a) Explain RSA digital signature algorithm with an example. (12)
   b) What is the limitation of existing internet system? Explain denial-of-service (DoS) attacks. (10)
   c) Define authentication. What are the threats to e-mail? Explain. (13)

## SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Make a comparison between copyrights and patents. (09)
   b) How trade secret protection is vanished by 'reverse engineering'? Explain. (08)
   c) How does operation system offer protection at several levels? (08)
   d) Explain paging combined with segmentation. What are the advantages and disadvantages of this technique for protecting in general purpose operating system? (10)

6. a) "Information, Security, and Control are three main ingredients of a secure system" – justify the statement. (09)
   b) What are the complementary goals of information security? Describe them vividly. (10)
   c) What is a threat in a computing system? Describe different types of threats with proper example(s). (10)
   d) Differentiate between AES and DES. (06)

7. a) What is data security? How does it differ from network security? (08)
   b) What is buffer overflow problem? How it affects in security implication of a system? (08)
   c) Differentiate between virus and worm. (06)
   d) What is virus gain control? Explain with proper figure(s). (08)
   e) How a virus can be identified? (05)



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8. a) What is directory system? Describe its advantages and disadvantages with appropriate figure(s). (12)
   b) How does military security model fulfill need–to–know rule and security requirements? (14)
   c) What is lattice? How is military security model represented by lattice? (09)

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